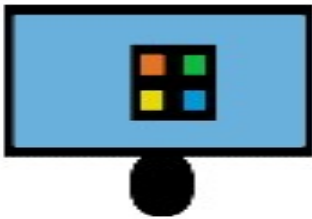


UD4 – Tarea Barco



Creado: Por mi

Imagen de fondo creada por IA, DALL-E

Creado por: Oscar Ruiz Bejarano

28/03/2025



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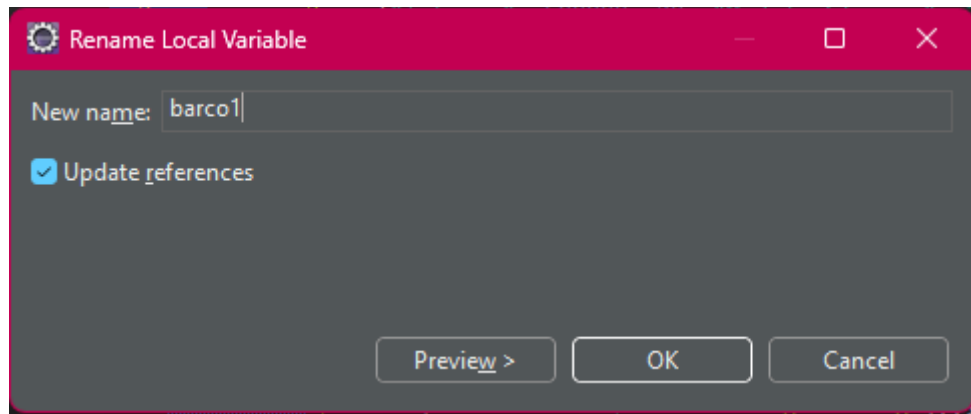
1. Introducción

En este trabajo he realizado una refactorización , utilizado PMD y generación de JavaDoc. Todo esto se ve reflejado en Git subiendo todas las cosas a mi repositorio remoto.

(Nota: Este documento se subirá al repositorio remoto también)

2. Refactorización

Seleccionamos el nombre de una variable que queramos cambiar y buscamos la opción de refactorización.



Refactorizado:

```
public static void main(String[] args) {
    Barco barco1;
    int stockActual;

    // Creación de un objeto Barco con modelo y color
    barco1 = new Barco("Titanic", 500000, 20, "Modelo Clásico", "Blanco");

    try {
        System.out.println("Venta de Barcos");
        barco1.vender(5);
    } catch (Exception e) {
        System.out.println("Fallo al vender: " + e.getMessage());
    }

    try {
        System.out.println("Compra de Barcos");
        barco1.comprar(10);
    } catch (Exception e) {
        System.out.println("Fallo al comprar: " + e.getMessage());
    }

    stockActual = barco1.obtenerStock();
    System.out.println("El stock actual es: " + stockActual);

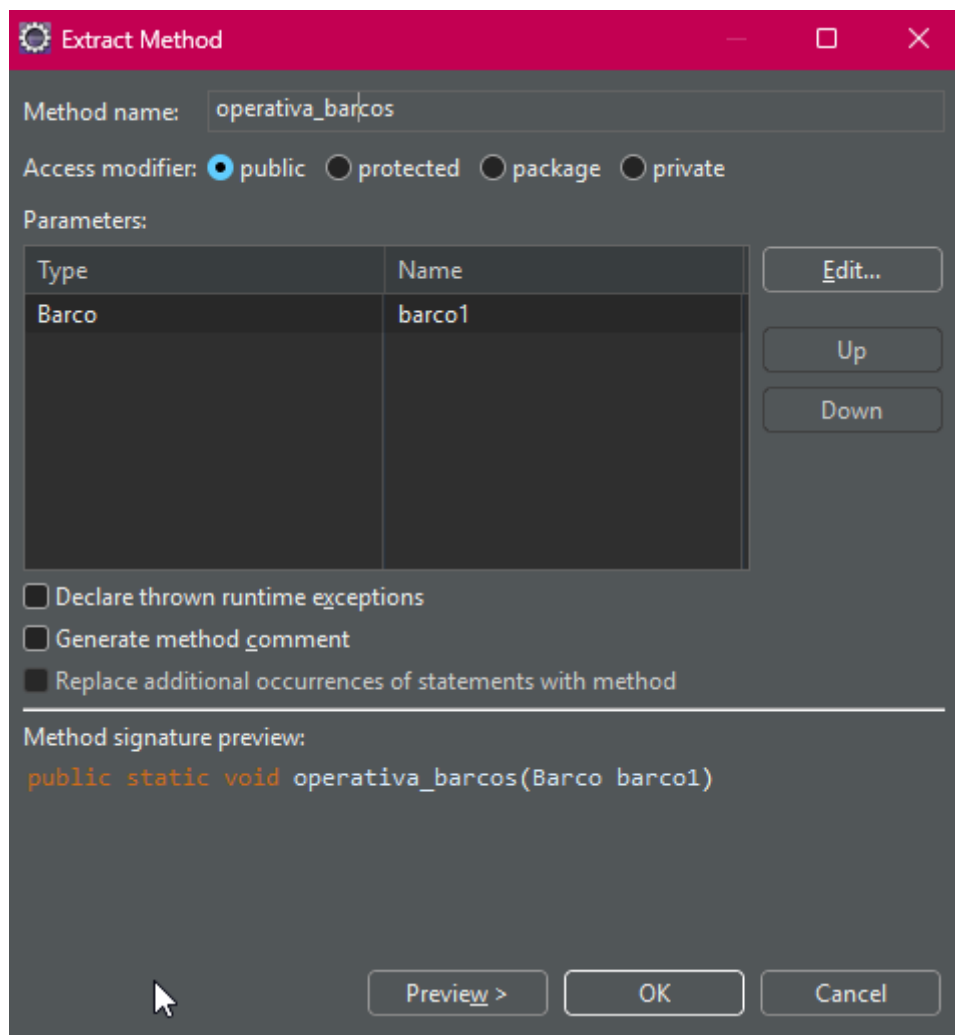
    double valorStock = barco1.calcularValorStock();
    System.out.println("El valor total del stock es: " + valorStock);
}
```

2.1 Añadir método operativa barcos

Seleccionamos las líneas que queremos generar el método:

```
try {  
    System.out.println("Venta de Barcos");  
    barco1.vender(5);  
} catch (Exception e) {  
    System.out.println("Fallo al vender: " + e.getMessage());  
}  
  
try {  
    System.out.println("Compra de Barcos");  
    barco1.comprar(10);  
} catch (Exception e) {  
    System.out.println("Fallo al comprar: " + e.getMessage());  
}
```

Una vez hecho eso vamos al menú de **refactorizar** y buscamos la opción de extraer método:



Le cambiamos el nombre y ponemos el método como **public**.

```
public static void operativa_barcos(Barco barco1) {
    try {
        System.out.println("Venta de Barcos");
        barco1.vender(5);
    } catch (Exception e) {
        System.out.println("Fallo al vender: " + e.getMessage());
    }

    try {
        System.out.println("Compra de Barcos");
        barco1.comprar(10);
    } catch (Exception e) {
        System.out.println("Fallo al comprar: " + e.getMessage());
    }
}
```

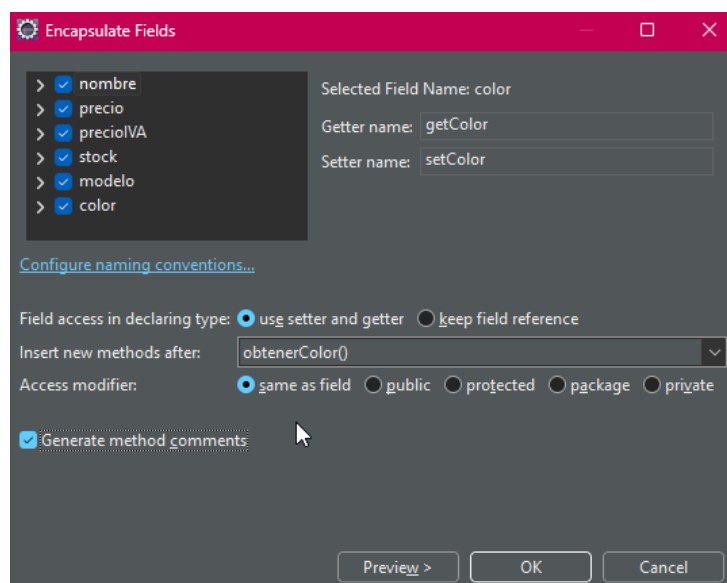
Se nos genera el método.

2.2. Encapsular campos

Dentro de la clase *Barco* seleccionamos las variables que queremos encapsular:

```
private String nombre;
private double precio;
private double precioIVA;
private int stock;
private String modelo;
private String color;
```

Vamos a **Refactor > Encapsulate Field**.



Seleccionamos todos los que queramos generar.

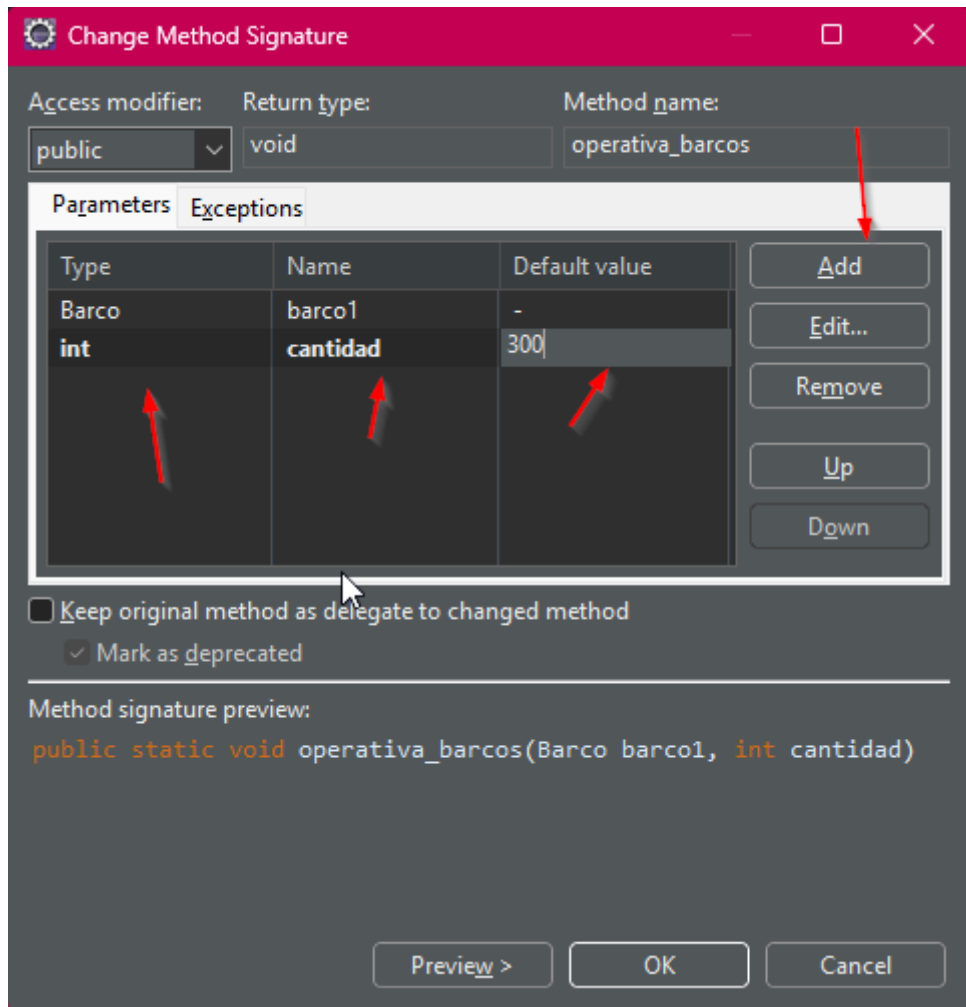
```
7- /**
8-  * @return the color
9-  */
10- private String getColor() {
11-     return color;
12- }
13-
14- /**
15-  * @param color the color to set
16-  */
17- private void setColor(String color) {
18-     this.color = color;
19- }
20-
21- /**
22-  * @return the modelo
23-  */
24- private String getModelo() {
25-     return modelo;
26- }
27-
28- /**
29-  * @param modelo the modelo to set
30-  */
31- private void setModelo(String modelo) {
32-     this.modelo = modelo;
33- }
34-
35- /**
36-  * @return the stock
37-  */
38- private int getStock() {
39-     return stock;
40- }
```

Se generan con sus comentarios.

2.3. Añadir parámetro cantidad a operativa barcos

Buscamos operativa_barcos en el main y luego vamos a **Refactor > Change Method Signature**.

Añadimos el parámetro cantidad con valor por defecto 300:



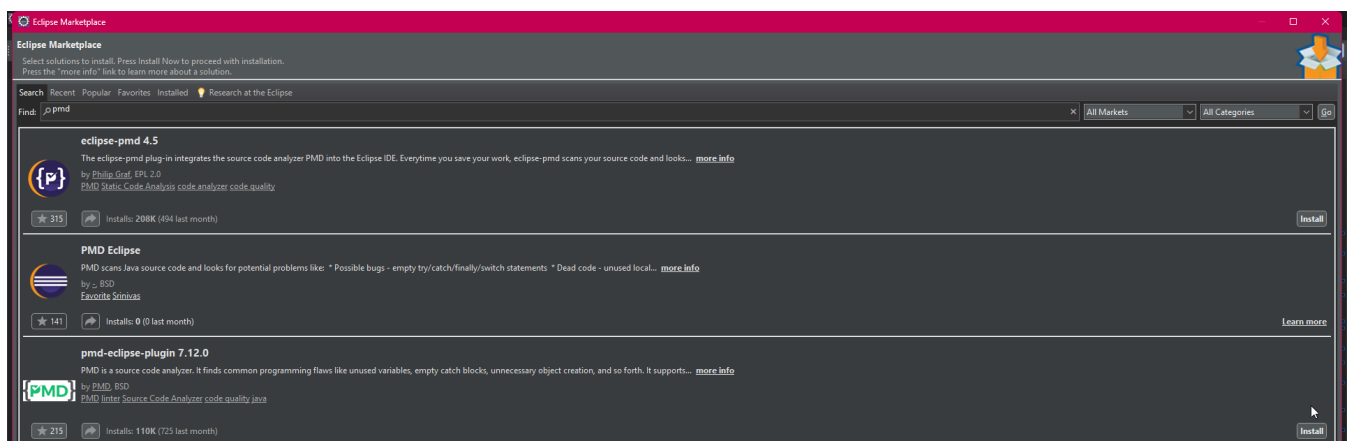
Como se puede ver se ha añadido:

```
public static void operativa_barcos(Barco barco1, int cantidad) {
    try {
        System.out.println("Venta de Barcos");
        barco1.vender(5);
    } catch (Exception e) {
        System.out.println("Fallo al vender: " + e.getMessage());
    }

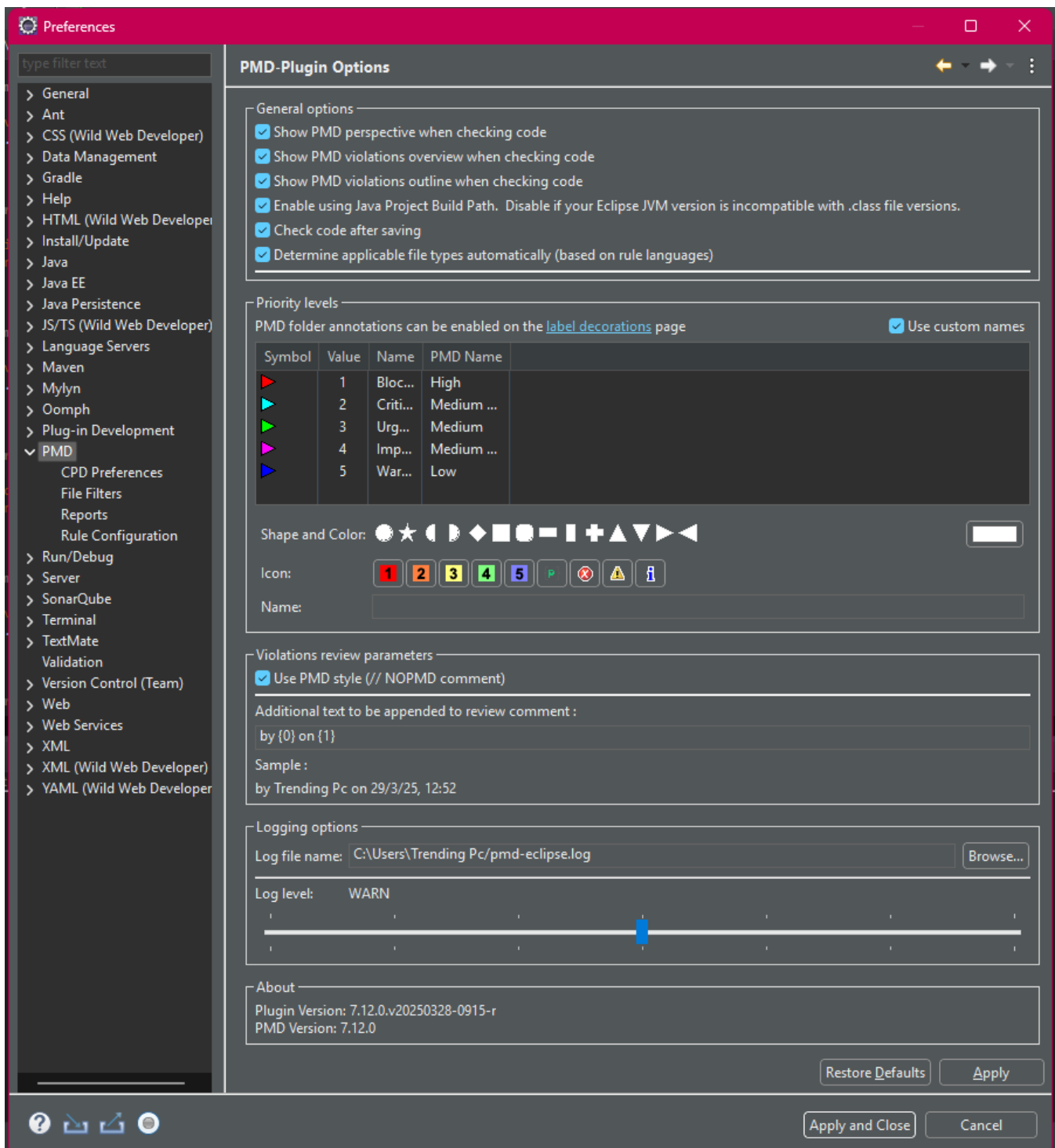
    try {
        System.out.println("Compra de Barcos");
        barco1.comprar(10);
    } catch (Exception e) {
        System.out.println("Fallo al comprar: " + e.getMessage());
    }
}
```

3. Analizador de código pmd

Tenemos que entrar a la **Marketplace** de Eclipse y buscar *PMD*.

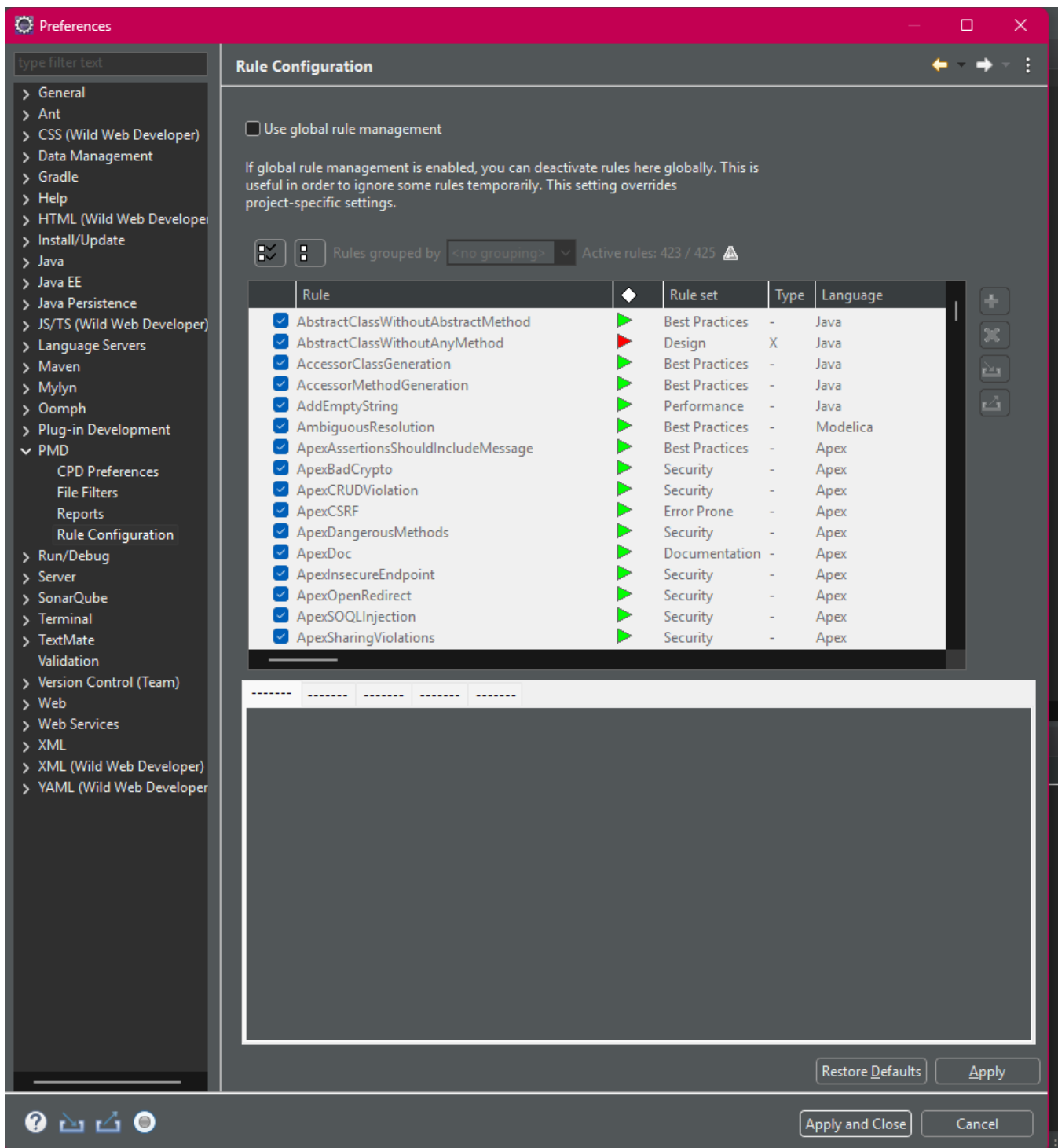


Vamos a **Windows > Preferences** y buscamos PMD.



Lo marcamos todo y le damos a aplicar.

Podemos cambiar las reglas:



Una vez lo tenemos instalado, analizamos el código:

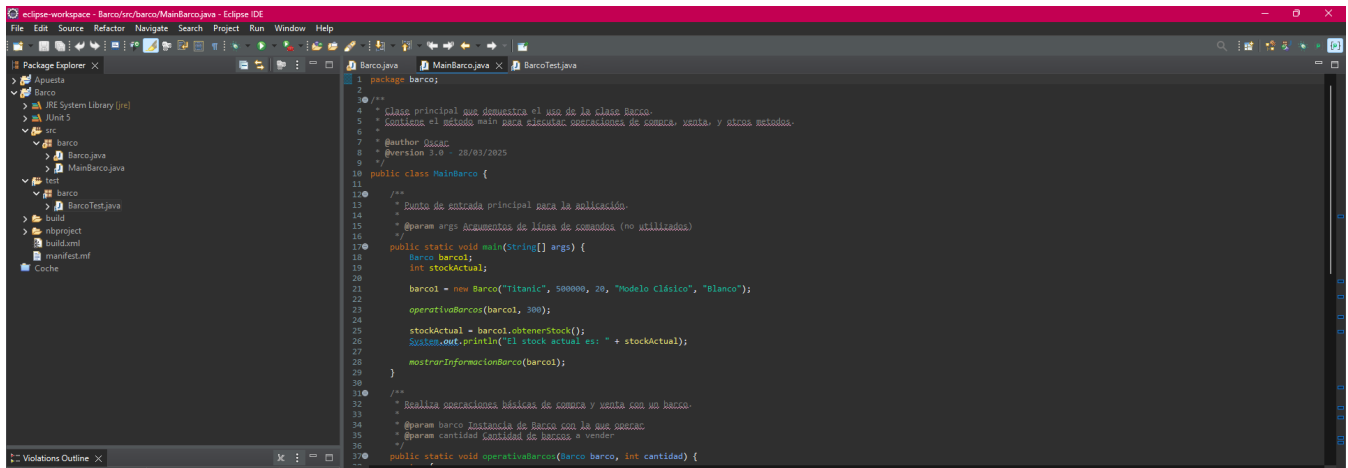
oi	Line	created	Rule	Error Message
▶	38	Sat Mar 29 13:08:59 ...	MethodNamingConventions	MethodNamingConventions
▶	34	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	50	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	28	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	21	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	43	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	24	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	47	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	30	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	35	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	40	Sat Mar 29 13:08:59 ...	SystemPrintln	SystemPrintln
▶	23	Sat Mar 29 13:08:59 ...	LocalVariableCouldBeFinal	LocalVariableCouldBeFinal
▶	11	Sat Mar 29 13:08:59 ...	CommentRequired	CommentRequired
▶	42	Sat Mar 29 13:08:59 ...	AvoidCatchingGenericException	AvoidCatchingGenericException
▶	12	Sat Mar 29 13:08:59 ...	LocalVariableCouldBeFinal	LocalVariableCouldBeFinal
▶	16	Sat Mar 29 13:08:59 ...	UseUnderscoresInNumericLiteral	UseUnderscoresInNumericLiteral

Este es el de la clase **Main**, vamos a analizar el código de la clase **Barco**.

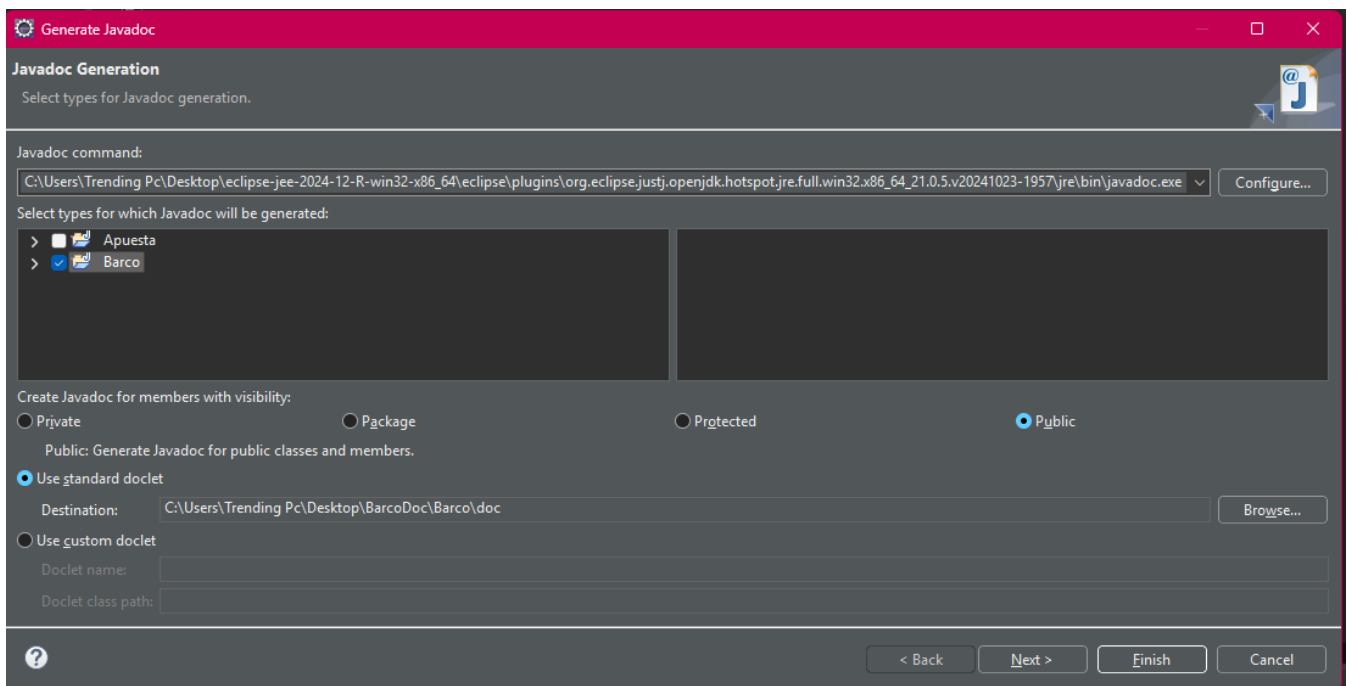
oi	Line	created	Rule	Error Message
▶	55	Sat Mar 29 13:11:44 ...	AvoidThrowingRawExceptionType	AvoidThrowingRawExceptionType
▶	69	Sat Mar 29 13:11:44 ...	AvoidThrowingRawExceptionType	AvoidThrowingRawExceptionType
▶	57	Sat Mar 29 13:11:44 ...	AvoidThrowingRawExceptionType	AvoidThrowingRawExceptionType
▶	49	Sat Mar 29 13:11:44 ...	AvoidThrowingRawExceptionType	AvoidThrowingRawExceptionType
▶	21	Sat Mar 29 13:11:44 ...	MethodArgumentCouldBeFinal	MethodArgumentCouldBeFinal
▶	139	Sat Mar 29 13:11:44 ...	UnusedPrivateMethod	UnusedPrivateMethod
▶	53	Sat Mar 29 13:11:44 ...	SignatureDeclareThrowsException	SignatureDeclareThrowsException
▶	35	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired
▶	74	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired
▶	10	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired
▶	21	Sat Mar 29 13:11:44 ...	MethodArgumentCouldBeFinal	MethodArgumentCouldBeFinal
▶	62	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired
▶	97	Sat Mar 29 13:11:44 ...	MethodArgumentCouldBeFinal	MethodArgumentCouldBeFinal
▶	132	Sat Mar 29 13:11:44 ...	UnusedPrivateMethod	UnusedPrivateMethod
▶	40	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired
▶	11	Sat Mar 29 13:11:44 ...	CommentRequired	CommentRequired

4. Documentación

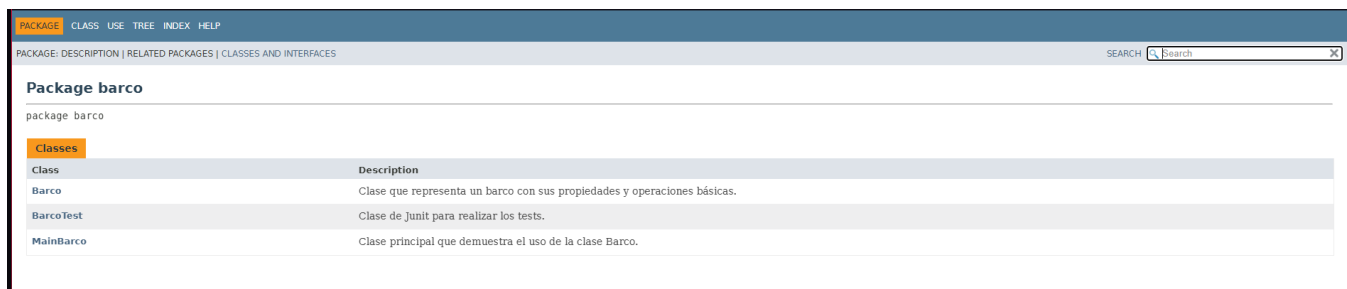
Tenemos que añadir comentarios para el java doc en todo los archivos del documento, lo vamos a hacer y luego generamos el *javadoc*.



Vamos a **Project > Generate JavaDoc**, pulsamos y seleccionamos que queremos generar.



Una vez generado se nos mostrará en nuestro navegador.



5. Git y GitHub

Tenemos que crear una carpeta y movernos en Git a esa carpeta:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~
$ cd Desktop/

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop
$ cd miPracticaBarco/
```

Una vez hecho eso, tenemos que realizar la configuración inicial:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco
$ git init
Initialized empty Git repository in C:/Users/Trending Pc/Desktop/miPracticaBarco/.git/

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git config user.name "Gim-Gim-Gim"

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git config user.email "gim-gim-gim@miPracticaBarco.com"
```

Tenemos que iniciar el repositorio y tenemos que configurar la cuenta de usuario y el email.

Una vez realizado el paso anterior, tenemos que configurar nuestro repositorio remoto (GitHub).

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git remote add origin https://github.com/Gim-Gim-Gim/miPracticaBarco.git
```

Miramos el *status* y hacemos un *add* y *commit*, para subir y confirmar la subida de los archivos disponibles en nuestro repositorio.

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git add .

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git commit -m "Primer commit: Código inicial del proyecto Barco"
On branch master

Initial commit

nothing to commit (create/copy files and use "git add" to track)
```

Vamos a crear un archivo de prueba para probar que todo funciona, repetimos el mismo proceso anterior:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ touch hola.txt

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hola.txt

nothing added to commit but untracked files present (use "git add" to track)

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git add hola.txt

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git status
On branch master

No commits yet

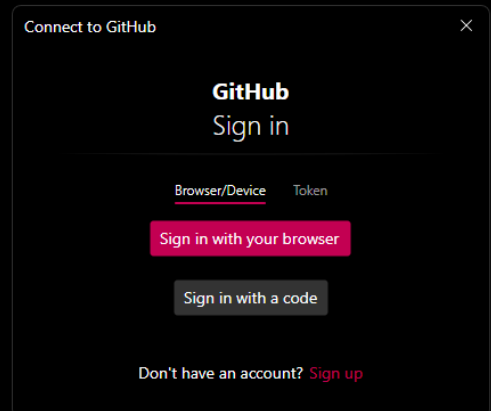
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   hola.txt
```

Por último, podemos ver que el archivo está listo para ser confirmado, por lo que ahora tenemos que realizar un *commit*.

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git commit -m "Realizo un commit"
[master (root-commit) f9e7873] Realizo un commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hola.txt
```

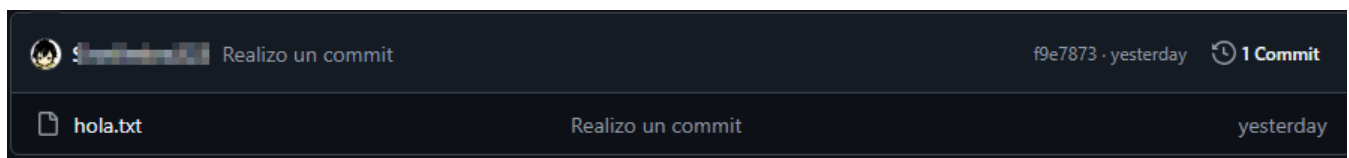
El siguiente paso es subir nuestra rama master a nuestro repositorio remoto.

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin master
```



Realizamos el *push -u* a *master* para que se nos suba.

Ahora toca ver si se ha subido la rama *master* con su contenido:



Una vez realizado todo eso, vamos a crear las distintas ramas para poder subir todos nuestros archivos del proyecto a nuestro repositorio.

Primero vamos a crear 5 ramas distintas para poder subir las 4 versiones del proyecto: codigoJavaOrigen, codigoJavaRefactorizado, modificacionesComunes, codigoDoc codigoAnalizado.

Creamos todas las ramas con *git branch nombre_rama*

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git branch
codigoAnalizado
codigoDoc
codigoJavaOrigen
codigoJavaRefactorizado
* master
modificacionesComunes
```

Tenemos que subir nuestra rama a nuestro repositorio:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push origin codigoJavaOrigen
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'codigoJavaOrigen' on GitHub by visiting:
remote:   https://github.com/SilverDove122/miPracticaBarco/pull/new/codigoJavaOrigen
remote:
To https://github.com/SilverDove122/miPracticaBarco.git
 * [new branch]      codigoJavaOrigen -> codigoJavaOrigen
```

Cambiamos de rama a la de código origen

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git checkout codigoJavaOrigen
Switched to branch 'codigoJavaOrigen'
Your branch is up to date with 'origin/codigoJavaOrigen'.
```


Subimos todos los archivos:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaOrigen)
$ git add .
warning: in the working copy of 'BarcoOriginal/BarcoNormal/src/barco/Barco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoOriginal/BarcoNormal/src/barco/MainBarco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoOriginal/BarcoNormal/test/barco/BarcoTest.java', LF will be replaced by CRLF the next time Git touches it
```

Confirmamos los cambios:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaOrigen)
$ git commit -m "He añadido el código original del proyecto"
[codigoJavaOrigen 1a21cfd] He añadido el código original del proyecto
21 files changed, 1477 insertions(+)
create mode 100644 BarcoOriginal/BarcoNormal/.classpath
create mode 100644 BarcoOriginal/BarcoNormal/.project
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/Barco.class
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/BarcoTest.class
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/MainBarco.class
create mode 100644 BarcoOriginal/BarcoNormal/build.xml
rename a.txt => BarcoOriginal/BarcoNormal/build/classes/.netbeans_automatic_build (100%)
create mode 100644 BarcoOriginal/BarcoNormal/build/classes/coche/Coche.class
create mode 100644 BarcoOriginal/BarcoNormal/build/classes/coche/Main.class
create mode 100644 BarcoOriginal/BarcoNormal/build/test/classes/.netbeans_automatic_build
create mode 100644 BarcoOriginal/BarcoNormal/build/test/classes/coche/CocheTest.class
create mode 100644 BarcoOriginal/BarcoNormal/manifest.mf
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/build-impl.xml
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/genfiles.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/private/private.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/private/private.xml
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/project.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/project.xml
create mode 100644 BarcoOriginal/BarcoNormal/src/barco/Barco.java
create mode 100644 BarcoOriginal/BarcoNormal/src/barco/MainBarco.java
create mode 100644 BarcoOriginal/BarcoNormal/test/barco/BarcoTest.java
```

Una vez hecho eso lo subimos (si es la primera vez que hacemos un push, hay que usar el push -u):

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaOrigen)
$ git push
Enumerating objects: 40, done.
Counting objects: 100% (40/40), done.
Delta compression using up to 16 threads
Compressing objects: 100% (32/32), done.
Writing objects: 100% (39/39), 20.17 KiB | 2.88 MiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sinn0mbre222/miPracticaBarco.git
f9e7873..1a21cfd  codigoJavaOrigen -> codigoJavaOrigen
```

Tenemos que repetir el mismo procedimiento para todas las demás ramas.

```
MINGW64/c/Users/Trending Pc/Desktop/miPracticaBarco

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git branch
  codigoAnalizado
  codigoDoc
  codigoJavaOrigen
  codigoJavaRefactorizado
* master
  modificacionesComunes

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin codigoAnalizado
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'codigoAnalizado' on GitHub by visiting:
remote:   https://github.com/Sinn0mbre222/miPracticaBarco/pull/new/codigoAnalizado
remote:
To https://github.com/Sinn0mbre222/miPracticaBarco.git
 * [new branch]      codigoAnalizado -> codigoAnalizado
branch 'codigoAnalizado' set up to track 'origin/codigoAnalizado'.

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin codigoDoc
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'codigoDoc' on GitHub by visiting:
remote:   https://github.com/Sinn0mbre222/miPracticaBarco/pull/new/codigoDoc
remote:
To https://github.com/Sinn0mbre222/miPracticaBarco.git
 * [new branch]      codigoDoc -> codigoDoc
branch 'codigoDoc' set up to track 'origin/codigoDoc'.

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin codigoJavaRefactorizado
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'codigoJavaRefactorizado' on GitHub by visiting:
remote:   https://github.com/Sinn0mbre222/miPracticaBarco/pull/new/codigoJavaRefactorizado
remote:
To https://github.com/Sinn0mbre222/miPracticaBarco.git
 * [new branch]      codigoJavaRefactorizado -> codigoJavaRefactorizado
branch 'codigoJavaRefactorizado' set up to track 'origin/codigoJavaRefactorizado'.

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin modificacionesComunes
branch 'modificacionesComunes' set up to track 'origin/modificacionesComunes'.
Everything up-to-date
```

Código refactorizado:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaRefactorizado)
$ git add .
warning: in the working copy of 'BarcoRefac/BarcoRefac/src/barco/Barco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoRefac/BarcoRefac/src/barco/MainBarco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoRefac/BarcoRefac/test/barco/BarcoTest.java', LF will be replaced by CRLF the next time Git touches it

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaRefactorizado)
$ git commit -m "He añadido el codigo refactorizado"
[codigoJavaRefactorizado f82d890] He añadido el codigo refactorizado
21 files changed, 1572 insertions(+)
create mode 100644 BarcoRefac/BarcoRefac/.classpath
create mode 100644 BarcoRefac/BarcoRefac/.project
create mode 100644 BarcoRefac/BarcoRefac/bin/barco/Barco.class
create mode 100644 BarcoRefac/BarcoRefac/bin/barco/BarcoTest.class
create mode 100644 BarcoRefac/BarcoRefac/bin/barco/MainBarco.class
create mode 100644 BarcoRefac/BarcoRefac/build.xml
create mode 100644 BarcoRefac/BarcoRefac/build/classes/.netbeans_automatic_build
create mode 100644 BarcoRefac/BarcoRefac/build/classes/coche/Coche.class
create mode 100644 BarcoRefac/BarcoRefac/build/classes/coche/Main.class
create mode 100644 BarcoRefac/BarcoRefac/build/test/classes/.netbeans_automatic_build
create mode 100644 BarcoRefac/BarcoRefac/build/test/classes/coche/CocheTest.class
create mode 100644 BarcoRefac/BarcoRefac/manifest.mf
create mode 100644 BarcoRefac/BarcoRefac/nbproject/build-impl.xml
create mode 100644 BarcoRefac/BarcoRefac/nbproject/genfiles.properties
create mode 100644 BarcoRefac/BarcoRefac/nbproject/private/private.properties
create mode 100644 BarcoRefac/BarcoRefac/nbproject/private/private.xml
create mode 100644 BarcoRefac/BarcoRefac/nbproject/project.properties
create mode 100644 BarcoRefac/BarcoRefac/nbproject/project.xml
create mode 100644 BarcoRefac/BarcoRefac/src/barco/Barco.java
create mode 100644 BarcoRefac/BarcoRefac/src/barco/MainBarco.java
create mode 100644 BarcoRefac/BarcoRefac/test/barco/BarcoTest.java

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoJavaRefactorizado)
$ git push
Enumerating objects: 38, done.
Counting objects: 100% (38/38), done.
Delta compression using up to 16 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (37/37), 20.68 KiB | 2.95 MiB/s, done.
Total 37 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sinn0mbre222/miPracticaBarco.git
f9e7873..f82d890 codigoJavaRefactorizado -> codigoJavaRefactorizado
```

Ahora hacemos lo mismo para el codigoAnalizado donde se verán las imágenes del código analizado:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoAnalizado)
$ git add .

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (codigoAnalizado)
$ git commit -m "he añadido las imagenes de codigo analizado"
[codigoAnalizado 77cf3ff] he añadido las imagenes de codigo analizado
7 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoMainOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoTestOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacBarco.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacMain.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacTest.png
delete mode 100644 a.txt
```

Por último hacemos lo mismo para el codigoDoc:

```

MINGW64 ~/Desktop/miPracticaBarco
$ git add
warning: in the working copy of 'BarcoDoc/BarcoDoc/doc/legal/jquery.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/doc/legal/jqueryUI.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/doc/script-dir/jquery-3.7.1.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/doc/script-dir/jquery-ui.min.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/doc/script-dir/jquery-ui.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/src/barco/Barco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/src/barco/MainBarco.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'BarcoDoc/BarcoDoc/test/barco/BarcoTest.java', LF will be replaced by CRLF the next time Git touches it
$ git commit -m "He añadido el código con el doc"
[codigoDoc 0075c90] He añadido el código con el doc
68 files changed, 6561 insertions(+)
create mode 100644 BarcoDoc/Barco/.project
create mode 100644 BarcoDoc/Barco/bin/.project
create mode 100644 BarcoDoc/BarcoDoc/.classpath
create mode 100644 BarcoDoc/BarcoDoc/.project
create mode 100644 BarcoDoc/BarcoDoc/bin/barco/Barco.class
create mode 100644 BarcoDoc/BarcoDoc/bin/barco/BarcoTest.class
create mode 100644 BarcoDoc/BarcoDoc/bin/barco/MainBarco.class
create mode 100644 BarcoDoc/BarcoDoc/build.xml
create mode 100644 BarcoDoc/BarcoDoc/build/classes/.netbeans_automatic_build
create mode 100644 BarcoDoc/BarcoDoc/build/classes/coche/Coche.class
create mode 100644 BarcoDoc/BarcoDoc/build/classes/coche/Main.class
create mode 100644 BarcoDoc/BarcoDoc/build/test/classes/.netbeans_automatic_build
create mode 100644 BarcoDoc/BarcoDoc/build/test/classes/coche/CocheTest.class
create mode 100644 BarcoDoc/BarcoDoc/doc/allclasses-index.html
create mode 100644 BarcoDoc/BarcoDoc/doc/allpackages-index.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/Barco.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/BarcoTest.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/MainBarco.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/class-use/Barco.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/class-use/BarcoTest.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/class-use/MainBarco.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/package-summary.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/package-tree.html
create mode 100644 BarcoDoc/BarcoDoc/doc/barco/package-use.html
create mode 100644 BarcoDoc/BarcoDoc/doc/copy.svg
create mode 100644 BarcoDoc/BarcoDoc/doc/element-list
create mode 100644 BarcoDoc/BarcoDoc/doc/help-doc.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-1.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-2.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-3.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-4.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-5.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-6.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-7.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index-files/index-8.html
create mode 100644 BarcoDoc/BarcoDoc/doc/index.html
create mode 100644 BarcoDoc/BarcoDoc/doc/legal/ADDITIONAL_LICENSE_INFO

```

Ahora crearemos una rama donde se ajuntan todas las carpetas con todo:

```

usuario@DESKTOP-3LHBG1B MINGW64 ~/Desktop/miPracticaBarco (master)
$ git branch trabajoCompleto

usuario@DESKTOP-3LHBG1B MINGW64 ~/Desktop/miPracticaBarco (master)
$ git branch
  codigoAnalizado
  codigoDoc
  codigoJavaOrigen
  codigoJavaRefactorizado
* master
  modificacionesComunes
  trabajoCompleto

usuario@DESKTOP-3LHBG1B MINGW64 ~/Desktop/miPracticaBarco (master)
$ git push -u origin trabajoCompleto
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'trabajoCompleto' on GitHub by visiting:
remote:   https://github.com/SinnOmbre222/miPracticaBarco/pull/new/trabajoCompleto
remote:
To https://github.com/SinnOmbre222/miPracticaBarco.git
 * [new branch]   trabajoCompleto -> trabajoCompleto
branch 'trabajoCompleto' set up to track 'origin/trabajoCompleto'.

```

Tenemos que hacer un *git merge* a todas las ramas.

Vamos hacer una por una:

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (trabajoCompleto)
$ git merge codigoJavaOrigen trabajoCompleto
Updating f9e7873..1a21cfd
Fast-forward
BarcoOriginal/BarcoNormal/.classpath | 12 +
BarcoOriginal/BarcoNormal/.project | 17 +
BarcoOriginal/BarcoNormal/bin/barco/Barco.class | Bin 0 -> 2132 bytes
.../BarcoNormal/bin/barco/BarcoTest.class | Bin 0 -> 2260 bytes
.../BarcoNormal/bin/barco/MainBarco.class | Bin 0 -> 2402 bytes
BarcoOriginal/BarcoNormal/build.xml | 74 ++
.../build/classes/.netbeans_automatic_build | 0
.../BarcoNormal/build/classes/coche/Coche.class | Bin 0 -> 1373 bytes
.../BarcoNormal/build/classes/coche/Main.class | Bin 0 -> 1322 bytes
.../build/test/classes/.netbeans_automatic_build | 0
.../build/test/classes/coche/CocheTest.class | Bin 0 -> 1978 bytes
BarcoOriginal/BarcoNormal/manifest.mf | 3 +
BarcoOriginal/BarcoNormal/nbproject/build-impl.xml | 1042 +++++
.../BarcoNormal/nbproject/genfiles.properties | 8 +
.../nbproject/private/private.properties | 2 +
.../BarcoNormal/nbproject/private/private.xml | 4 +
.../BarcoNormal/nbproject/project.properties | 72 ++
BarcoOriginal/BarcoNormal/nbproject/project.xml | 15 +
BarcoOriginal/BarcoNormal/src/barco/Barco.java | 86 ++
BarcoOriginal/BarcoNormal/src/barco/MainBarco.java | 49 +
.../BarcoNormal/test/barco/BarcoTest.java | 93 ++
21 files changed, 1477 insertions(+)
create mode 100644 BarcoOriginal/BarcoNormal/.classpath
create mode 100644 BarcoOriginal/BarcoNormal/.project
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/Barco.class
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/BarcoTest.class
create mode 100644 BarcoOriginal/BarcoNormal/bin/barco/MainBarco.class
create mode 100644 BarcoOriginal/BarcoNormal/build.xml
create mode 100644 BarcoOriginal/BarcoNormal/build/classes/.netbeans_automatic_build
create mode 100644 BarcoOriginal/BarcoNormal/build/classes/coche/Coche.class
create mode 100644 BarcoOriginal/BarcoNormal/build/classes/coche/Main.class
create mode 100644 BarcoOriginal/BarcoNormal/build/test/classes/.netbeans_automatic_build
create mode 100644 BarcoOriginal/BarcoNormal/build/test/classes/coche/CocheTest.class
create mode 100644 BarcoOriginal/BarcoNormal/manifest.mf
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/build-impl.xml
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/genfiles.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/private/private.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/private/private.xml
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/project.properties
create mode 100644 BarcoOriginal/BarcoNormal/nbproject/project.xml
create mode 100644 BarcoOriginal/BarcoNormal/src/barco/Barco.java
create mode 100644 BarcoOriginal/BarcoNormal/src/barco/MainBarco.java
create mode 100644 BarcoOriginal/BarcoNormal/test/barco/BarcoTest.java
```


Código refactorizado:

```

Merge made by the 'ort' strategy.
BarcoRefact/BarcoRefac/.classpath | 12 +
BarcoRefact/BarcoRefac/.project | 17 +
BarcoRefact/BarcoRefac/bin/barco/Barco.class | Bin 0 -> 3210 bytes
BarcoRefact/BarcoRefac/bin/barco/BarcoTest.class | Bin 0 -> 2260 bytes
BarcoRefact/BarcoRefac/bin/barco/MainBarco.class | Bin 0 -> 2535 bytes
BarcoRefact/BarcoRefac/build.xml | 74 ++
.../build/classes/.netbeans_automatic_build | 0
.../BarcoRefac/build/classes/coche/Coche.class | Bin 0 -> 1373 bytes
.../BarcoRefac/build/classes/coche/Main.class | Bin 0 -> 1322 bytes
.../build/test/classes/.netbeans_automatic_build | 0
.../build/test/classes/coche/CocheTest.class | Bin 0 -> 1978 bytes
BarcoRefact/BarcoRefac/manifest.mf | 3 +
BarcoRefact/BarcoRefac/nbproject/build-impl.xml | 1042 +++++
.../BarcoRefac/nbproject/genfiles.properties | 8 +
.../nbproject/private/private.properties | 2 +
.../BarcoRefac/nbproject/private/private.xml | 4 +
.../BarcoRefac/nbproject/project.properties | 72 ++
BarcoRefact/BarcoRefac/nbproject/project.xml | 15 +
BarcoRefact/BarcoRefac/src/barco/Barco.java | 171 +++++
BarcoRefact/BarcoRefac/src/barco/MainBarco.java | 56 ++
BarcoRefact/BarcoRefac/test/barco/BarcoTest.java | 96 ++
21 files changed, 1572 insertions(+)
create mode 100644 BarcoRefact/BarcoRefac/.classpath
create mode 100644 BarcoRefact/BarcoRefac/.project
create mode 100644 BarcoRefact/BarcoRefac/bin/barco/Barco.class
create mode 100644 BarcoRefact/BarcoRefac/bin/barco/BarcoTest.class
create mode 100644 BarcoRefact/BarcoRefac/bin/barco/MainBarco.class
create mode 100644 BarcoRefact/BarcoRefac/build.xml
create mode 100644 BarcoRefact/BarcoRefac/build/classes/.netbeans_automatic_build
create mode 100644 BarcoRefact/BarcoRefac/build/classes/coche/Coche.class
create mode 100644 BarcoRefact/BarcoRefac/build/classes/coche/Main.class
create mode 100644 BarcoRefact/BarcoRefac/build/test/classes/.netbeans_automatic_build
create mode 100644 BarcoRefact/BarcoRefac/build/test/classes/coche/CocheTest.class
create mode 100644 BarcoRefact/BarcoRefac/manifest.mf
create mode 100644 BarcoRefact/BarcoRefac/nbproject/build-impl.xml
create mode 100644 BarcoRefact/BarcoRefac/nbproject/genfiles.properties
create mode 100644 BarcoRefact/BarcoRefac/nbproject/private/private.properties
create mode 100644 BarcoRefact/BarcoRefac/nbproject/private/private.xml
create mode 100644 BarcoRefact/BarcoRefac/nbproject/project.properties
create mode 100644 BarcoRefact/BarcoRefac/nbproject/project.xml
create mode 100644 BarcoRefact/BarcoRefac/src/barco/Barco.java
create mode 100644 BarcoRefact/BarcoRefac/src/barco/MainBarco.java
create mode 100644 BarcoRefact/BarcoRefac/test/barco/BarcoTest.java

```

```

Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (trabajoCompleto)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 375 bytes | 375.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Sinn0mbre222/miPracticaBarco.git
1a21cfd..b0554d5 trabajoCompleto -> trabajoCompleto

```

Imágenes analizadas:

```
Merge made by the 'ort' strategy.
.../CodigoOrigen/CodigoBarcoMainOrigen.png | Bin 0 -> 107104 bytes
.../CodigoOrigen/CodigoBarcoOrigen.png | Bin 0 -> 137841 bytes
.../CodigoOrigen/CodigoBarcoTestOrigen.png | Bin 0 -> 164361 bytes
.../CodigoRefac/CodigoRefacBarco.png | Bin 0 -> 152061 bytes
.../CodigoRefac/CodigoRefacMain.png | Bin 0 -> 115698 bytes
.../CodigoRefac/CodigoRefacTest.png | Bin 0 -> 143069 bytes
6 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoMainOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoOrigen/CodigoBarcoTestOrigen.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacBarco.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacMain.png
create mode 100644 ImagenesCodigoAnalizado/CodigoRefac/CodigoRefacTest.png
```

```
Trending Pc@DESKTOP-QJTD007 MINGW64 ~/Desktop/miPracticaBarco (trabajoCompleto)
$ git push
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 16 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (13/13), 663.41 KiB | 28.84 MiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/SinnOmbre222/miPracticaBarco.git
b0554d5..ae6dcb0 trabajoCompleto -> trabajoCompleto
```

Codigo doc:

```
Merge made by the 'ort' strategy.
BarcoDoc/Barco/.project | 17 +
BarcoDoc/Barco/bin/.project | 17 +
BarcoDoc/BarcoDoc/.classpath | 12 +
BarcoDoc/BarcoDoc/.project | 17 +
BarcoDoc/BarcoDoc/bin/barco/Barco.class | Bin 0 -> 3210 bytes
BarcoDoc/BarcoDoc/bin/barco/BarcoTest.class | Bin 0 -> 2448 bytes
BarcoDoc/BarcoDoc/bin/barco/MainBarco.class | Bin 0 -> 2653 bytes
BarcoDoc/BarcoDoc/build.xml | 74 ++
.../build/classes/.netbeans_automatic_build | 0
BarcoDoc/BarcoDoc/build/classes/coche/Coche.class | Bin 0 -> 1373 bytes
BarcoDoc/BarcoDoc/build/classes/coche/Main.class | Bin 0 -> 1322 bytes
.../build/test/classes/.netbeans_automatic_build | 0
.../build/test/classes/coche/CocheTest.class | Bin 0 -> 1978 bytes
BarcoDoc/BarcoDoc/doc/allclasses-index.html | 76 ++
BarcoDoc/BarcoDoc/doc/allpackages-index.html | 64 +
BarcoDoc/BarcoDoc/doc/barco/Barco.html | 373 +++++
BarcoDoc/BarcoDoc/doc/barco/BarcoTest.html | 284 +++++
BarcoDoc/BarcoDoc/doc/barco/MainBarco.html | 222 +++++
BarcoDoc/BarcoDoc/doc/barco/class-use/Barco.html | 83 ++
.../BarcoDoc/doc/barco/class-use/BarcoTest.html | 57 +
.../BarcoDoc/doc/barco/class-use/MainBarco.html | 57 +
BarcoDoc/BarcoDoc/doc/barco/package-summary.html | 101 ++
BarcoDoc/BarcoDoc/doc/barco/package-tree.html | 77 ++
BarcoDoc/BarcoDoc/doc/barco/package-use.html | 74 ++
BarcoDoc/BarcoDoc/doc/copy.svg | 33 +
BarcoDoc/BarcoDoc/doc/element-list | 1 +
BarcoDoc/BarcoDoc/doc/help-doc.html | 183 +++
BarcoDoc/BarcoDoc/doc/index-files/index-1.html | 69 ++
BarcoDoc/BarcoDoc/doc/index-files/index-2.html | 83 ++
BarcoDoc/BarcoDoc/doc/index-files/index-3.html | 69 ++
BarcoDoc/BarcoDoc/doc/index-files/index-4.html | 75 ++
BarcoDoc/BarcoDoc/doc/index-files/index-5.html | 81 ++
BarcoDoc/BarcoDoc/doc/index-files/index-6.html | 65 +
BarcoDoc/BarcoDoc/doc/index-files/index-7.html | 81 ++
BarcoDoc/BarcoDoc/doc/index-files/index-8.html | 65 +
BarcoDoc/BarcoDoc/doc/index.html | 26 +
.../BarcoDoc/doc/legal/ADDITIONAL_LICENSE_INFO | 1 +
BarcoDoc/BarcoDoc/doc/legal/ASSEMBLY_EXCEPTION | 1 +
BarcoDoc/BarcoDoc/doc/legal/LICENSE | 1 +
BarcoDoc/BarcoDoc/doc/legal/jquery.md | 26 +
BarcoDoc/BarcoDoc/doc/legal/jqueryUI.md | 49 +
BarcoDoc/BarcoDoc/doc/link.svg | 31 +
BarcoDoc/BarcoDoc/doc/member-search-index.js | 1 +
BarcoDoc/BarcoDoc/doc/module-search-index.js | 1 +
BarcoDoc/BarcoDoc/doc/overview-tree.html | 81 ++
BarcoDoc/BarcoDoc/doc/package-search-index.js | 1 +
BarcoDoc/BarcoDoc/doc/resources/glass.png | Bin 0 -> 499 bytes
BarcoDoc/BarcoDoc/doc/resources/x.png | Bin 0 -> 394 bytes
.../BarcoDoc/doc/script-dir/jquery-3.7.1.min.js | 2 +
```

```
$ git push
Enumerating objects: 95, done.
Counting objects: 100% (95/95), done.
Delta compression using up to 16 threads
Compressing objects: 100% (80/80), done.
Writing objects: 100% (93/93), 103.33 KiB | 5.44 MiB/s, done.
Total 93 (delta 22), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (22/22), completed with 1 local object.
To https://github.com/SinnOmbre222/miPracticaBarco.git
   ae6dcb0..c7b12dd  trabajoCompleto -> trabajoCompleto
```

Comprobamos si se nos ha subido todo:

The screenshot shows the GitHub interface for the repository 'trabajoCompleto'. At the top, it indicates '7 Branches' and '0 Tags'. A status bar shows 'This branch is 9 commits ahead of master'. Below this, a commit history table is displayed with the following entries:

Commit Message	Commit Hash	Time Ago
Merge branch 'codigoDoc' into trabajoCompleto	c7b12dd	5 minutes ago
BarcoDoc		15 hours ago
BarcoOriginal/BarcoNormal		2 days ago
BarcoRefact/BarcoRefac		19 hours ago
ImagenesCodigoAnalizado		15 hours ago
hola.txt		3 days ago

Por último este documento se subirá también al trabajo completo.